Day Three Assignment: Answers

MIT 91205 Big Data Analytics

1. Schema-on-read means the data has no schema.

( ) True

(x) **False**

The schema is still there. Hive applies it when the data is read, not when it is loaded.

2. Why are columnar formats like Parquet/ORC preferred for analytical queries?

( ) a. They are row-by-row

(x) **b. They enable column projection and better compression**

( ) c. They require no compression

( ) d. They cannot be queried with SQL

A query reads only the columns it needs, and values from one column compress well together.

3. Partitioning and bucketing are the same thing.

( ) True

(x) **False**

Partitioning splits data into folders by value. Bucketing hashes rows into a fixed number of files.

4. Bucketing in Hive writes rows into a fixed number of files based on:

( ) a. File size

( ) b. Partition value

(x) **c. Hash of a column**

( ) d. Random assignment

The same value always lands in the same bucket file.

5. Which UDF type produces many output rows per input row (e.g., splitting a string)?

( ) a. Scalar UDF

( ) b. UDAF

( ) c. SerDe

(x) **d. UDTF**

One row in, many rows out, like explode().

6. Match each concept to its correct description.

Concept	**Matching description**
Pig Latin	Dataflow scripting language
UDTF	Produces many output rows per input row
Metastore	Central catalog of table and partition definitions
Managed table	Hive owns and drops the underlying data files

Managed or external comes down to who owns the files.

7. What is the role of a SerDe in Hive?

( ) a. Manages YARN containers

( ) b. Schedules jobs

( ) c. Replicates HDFS blocks

(x) **d. Converts between byte streams and rows**

SerDe is short for serializer and deserializer.

8. What does schema-on-read mean in Hive?

( ) a. Schema is applied when data is written
( ) b. Schema is stored in the data files only
(x) **c. Schema is applied when data is read**
( ) d. There is no schema

Traditional databases check the schema on write instead.

9. Which type of database is most commonly used as the backend for the Hive metastore?

Answer: **Relational database** (for example MySQL or PostgreSQL)

Derby, the built-in option, only supports one user, so it is used for testing.

10. Hive partitioning primarily enables:

( ) a. Encryption
(x) **b. Partition pruning (skip unrelated directories)**
( ) c. Replication of blocks
( ) d. YARN scheduling

It only helps when the query filters on the partition column.

11. Which statement about Hive's execution engine is correct?

(x) **a. Hive is a query layer that can use MapReduce, Tez, or Spark**
( ) b. Hive cannot query HDFS
( ) c. Hive has its own bespoke engine
( ) d. Hive is always MapReduce

The engine is chosen with hive.execution.engine.

12. Which Pig complex type is an unordered collection of tuples?

(x) **a. Bag**
( ) b. Scalar
( ) c. Map
( ) d. Tuple

A tuple is ordered; a bag is an unordered group of tuples.

13. Which statement best describes Pig Latin's programming style?

( ) a. Declarative, SQL-like
( ) b. Functional-only
( ) c. Object-oriented
(x) **d. Procedural, dataflow-oriented**

Pig is written as steps; SQL only states the result you want.

14. What does the Hive metastore store?

( ) a. Query results
(x) **b. Table, column, partition metadata**
( ) c. YARN containers
( ) d. The physical data files

The data files themselves stay in HDFS.

15. What happens when you DROP an external Hive table?

( ) a. Both metadata and files are deleted
( ) b. Data files are deleted
(x) **c. Only metadata is removed; files stay**
( ) d. Nothing happens

Dropping a managed table would delete the files too.